

Subject Description Form

Subject Code	DSAI3102
Subject Title	Multivariate Statistical Analysis
Credit Value	3
Level	3
Pre-requisite/ Co-requisite/ Exclusion	Pre-requisite: AMA3640
Objectives	Upon completion, students will be able to: Master classical multivariate methods (e.g., PCA, Factor Analysis, Classification). Implement, visualize, and interpret results using R or Python. Communicate findings from complex, multidimensional datasets effectively.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a) Understand the changelings in multivariate data analysis. b) Master classical multivariate methods (e.g., PCA, Factor Analysis, Classification). c) Implement, visualize, and interpret results using R or Python. d) Communicate findings from complex, multidimensional datasets effectively.
Subject Synopsis/ Indicative Syllabus	1. Intro to Multivariate Data & Matrix Review; 2. The Multivariate Normal Distribution; 3. Principal Component Analysis (PCA); 4. Factor Analysis & Canonical Correlation (CCA); 5. Classification (LDA, QDA) & Logistic Regression; 6. Modern Clustering; 7. Dimensionality Reduction (t-SNE, UMAP); 8. Selected Advanced Topics.
Teaching/Learning Methodology	The subject will be delivered mainly through lectures and tutorials. The lectures will be conducted to introduce the statistical inference concepts in the syllabus, which are then reinforced by learning activities involving demonstration, tutorial exercise and assignments/quizzes.

Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)			
			a	b	c	d
	1. Assignments	25%	√	√	√	
	2. Course Project	25%	√	√	√	
	4. Final Exam	50%	√		√	√
	Total	100 %				
	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Assignments: Tests understanding of core principles (a), practical use of models (b), and implement and evaluate (c). Course Project: Assesses tool use and application development (b), model implementation and evaluation (c), and demonstration of foundational knowledge in the written report (a). Final Exam: Measures comprehensive knowledge of architectures and theory (a), ability to design practical solutions through data analysis (c), and critical awareness of model limitations and societal implications (d).					
Student Study Effort Expected	Class contact:					
	▪ Lectures				26 Hrs.	
	▪ Tutorials				13 Hrs.	
	Other student study effort:					
	▪ Assignments				25 Hrs.	
	▪ Course Project				38 Hrs.	
	▪ Exam Preparation/Self-Study/Practice				23 Hrs.	
	Total student study effort				125 Hrs.	
Reading List and References	Reference Materials: 1. Johnson, R. A., & Wichern, D. W. (2007). Applied Multivariate Statistical Analysis (6th ed.). Pearson. 2. Hastie, T., Tibshirani, R., & Friedman, J. (2009). The elements of statistical learning: Data mining, inference, and prediction (2nd ed.). Springer.					